ONE-PAGE BRIEF

Post-quantum cryptography review: coinbase-cb-mpc @ 0b716706f998

Client: Coinbase — public repository (passive review) · Source: github.com/coinbase/cb-mpc, commit `0b716706f998` · Reviewed: 2026-11-05 09:56 UTC · Risk profile: Crypto exchange, custodian or wallet

POST-QUANTUM MATURITY
Level 1 of 4
post-quantum code present, not wired

FINDINGS
0 Critical **0** High **11** Medium
0 Low **0** Info

QUANTUM TIMELINE (MOSCA: X + Y − Z)
Key establishment: 10 + 5 − 8 = 7 → **too late**
Signatures: 10 + 5 − 8 = 7 → **too late**

Next step: Wire the existing post-quantum primitive into a critical path (key wrapping or the channel) behind a hybrid construction, and test interoperability.

WHAT MATTERS MOST

Medium QSI-2026-14003	**Quantum-vulnerable ECDH: key wrapping and data encryption at rest (5 file(s))** confirmed by the analyst What to do: Wrap keys with ML-KEM or a hybrid KEM (X25519 + ML-KEM-768) and re-wrap the stored keys; data that may already have been copied needs its...
Medium QSI-2026-13998	**Quantum-vulnerable Paillier: threshold signing and MPC protocols (15 file(s))** confirmed by the analyst What to do: Track post-quantum threshold signatures (NIST IR 8214C) and keep the protocol layer ready to swap the signature scheme; record which keys...
Medium QSI-2026-14001	**Quantum-vulnerable Threshold ECDSA: threshold signing and MPC protocols (13 file(s))** confirmed by the analyst What to do: Track post-quantum threshold signatures (NIST IR 8214C) and keep the protocol layer ready to swap the signature scheme; record which keys...
Medium QSI-2026-14002	**Quantum-vulnerable Ed25519: threshold signing and MPC protocols (9 file(s))** confirmed by the analyst What to do: Track post-quantum threshold signatures (NIST IR 8214C) and keep the protocol layer ready to swap the signature scheme; record which keys...

7 more in the full report.

WHAT THE QUANTUM-VULNERABLE CRYPTOGRAPHY PROTECTS

Role	Algorithms	What to do
Medium key wrapping and data encryption at rest	ECDH	Wrap keys with ML-KEM or a hybrid KEM (X25519 + ML-KEM-768) and re-wrap the stored keys; data that may...
Medium threshold signing and MPC protocols	Paillier, Threshold ECDSA, Ed25519, Pedersen commitment ...	Track post-quantum threshold signatures (NIST IR 8214C) and keep the protocol layer ready to swap the...
Medium transaction and message signing	ECDSA	Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum...
Medium role not determined	ECDSA, Ed25519, Schnorr (BIP-340), RSA	Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205)...

Analyst review: 10 confirmed, 0 not applicable, 0 dismissed as false positives. (QSI review, 2026-10-28)

Static review of source code and manifests; files analysed: 174. No requests were sent to the owner's systems.

Full report, content SHA-256: `fdfb9330457c22987ea1c4cd303f83521adf2040315630276702ada9a3d35546`

Full report signature (report.sig.json): Ed25519 key, SHA-256 fingerprint `5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`. bundle.sig.json lists the SHA-256 of every file in this bundle and is signed too; verify-bundle checks it offline.

Preliminary findings need confirmation in the code; the full report gives the evidence, the limits of each check and how to re-test.